PharmaLeaf · B.Pharm Semester VI · PCI 2026 (NEP 2020)

BP601T Advanced Pharmacognosy

PharmaLeaf notes for B.Pharm Semester VI (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order. This course shows how **traditional knowledge and natural products become modern medicines**: reverse pharmacology, databases, metabolomics and network pharmacology, AI and in-silico tools, bioassay-guided isolation, lead optimisation, preclinical and clinical development, and the patent and biodiversity laws that protect natural products and traditional knowledge.

Syllabus at a glance

BP601T is a 3-credit core theory course: 3 lectures per week, 45 hours in total, 75 marks (30 sessional + 45 end-semester).

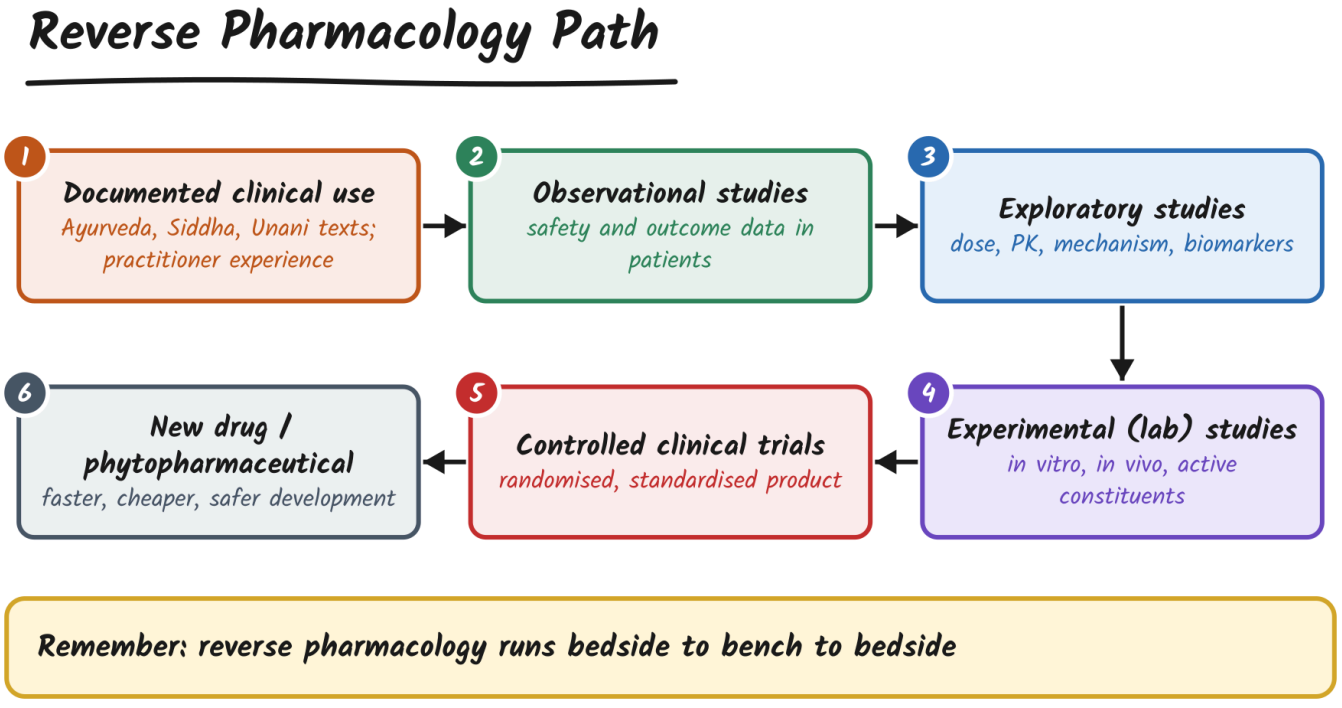
Unit	Title	Topics	Hours
I	Reverse pharmacology and integrative approaches	Reverse pharmacology (AYUSH view); ethnopharmacology and bioprospecting; traditional medicine databases; docking and ADMET screening; herbal adjuvant therapy in metabolic and non-communicable diseases	6
II	Metabolomics and systems biology	Metabolomics tools (NMR, HRMS); profiling and dereplication; quality control and validation of traditional claims; network pharmacology; spectral libraries and chemoinformatics databases	11
III	Modern techniques in natural product discovery	AI, machine learning and big data; docking, virtual screening, pharmacophore modelling; genomic and transcriptomic tools; high-throughput screening and bioautography	10
IV	Validation and development of herbal leads	Bioactivity-guided fractionation; structure elucidation; SAR and QSAR of salicin, artemisinin, piperine, papaverine and andrographolide derivatives; OECD toxicity testing; pharmacokinetics; clinical trials; IND, NDA, review and post-marketing surveillance	12
V	Patenting of natural products	Patent, IPR, farmers' and breeders' rights; bioprospecting and biopiracy; patenting natural substances and traditional knowledge; TKDL; National Biodiversity Authority; Nagoya Protocol; turmeric and neem case studies	6

Unit I: Reverse Pharmacology and Integrative Approaches (6 hours)

1.1 Reverse pharmacology

Classical (forward) drug discovery: laboratory to clinic *("bench to bedside"): target identification, then screening of thousands of compounds, lead optimisation, animal studies and finally clinical trials. It takes* ***10–15 years*** *and very large costs, and most candidates fail.*

Reverse pharmacology (RP): *the science of* ***integrating documented clinical and experiential observations*** *(from traditional medicine such as* ***Ayurveda****) into* ***leads****, through* ***transdisciplinary exploratory studies****, and developing them into drug candidates by* ***experimental and clinical research****. It goes "****bedside to bench to bedside****": it* ***starts from human use****, so safety and efficacy signals are already known. The concept was developed in India by* ***Dr Ashok D. B. Vaidya*** *and colleagues ("****the golden triangle****" of* ***Ayurveda, modern medicine and modern science****).*



Three phases of reverse pharmacology

Phase	What is done	Example
1. Experiential phase	Collect **documented clinical experience**: classical texts, **traditional practitioners**, patient records, safety of long human use; select the formulation or plant and the disease	Ayurvedic use of *Commiphora wightii* (guggulu) for "medoroga" (obesity and lipid disorders)
2. Exploratory phase	**Small clinical studies** (dose-finding, tolerability, pharmacodynamic markers) and **basic in-vitro and in-vivo studies** to understand the mechanism, standardise the material and identify active fractions	Lipid-lowering studies of guggul extract; isolation of guggulsterones
3. Experimental phase	**Detailed preclinical** (toxicology, pharmacology, mechanism, targets) and **controlled clinical trials**; development of standardised extract, phytopharmaceutical or pure lead	Guggulipid developed by **CSIR-CDRI**

Advantages of reverse pharmacology: lower cost and shorter time (human safety known), **higher success rate**, **lower risk of toxicity**, ethical starting point, scientific validation of traditional medicine, may yield **new mechanisms** (multi-target actions), affordable products for developing countries.

Limitations: variable traditional claims (different interpretations, diagnoses not matching modern disease names), complex multi-component mixtures, standardisation difficulties, possible **biopiracy** concerns, need for good-quality clinical trials.

Examples of drugs and products from reverse pharmacology / traditional knowledge

Product	Source and traditional basis	Outcome
Reserpine	*Rauvolfia serpentina* (sarpagandha): Ayurvedic use for insanity and hypertension	First modern antihypertensive and antipsychotic alkaloid (1950s)
Artemisinin	*Artemisia annua*: Chinese text (Ge Hong, 4th century) on fever	Antimalarial; Nobel Prize 2015 (Tu Youyou)
Guggulipid	Guggulu resin: Ayurvedic "medohara"	Lipid-lowering product (CSIR-CDRI)
Bacopa (Bacomind, Memory Plus)	Brahmi: medhya rasayana	Memory-enhancing standardised extracts
Picroliv	*Picrorhiza kurroa* (kutki)	Hepatoprotective iridoid fraction (CSIR-CDRI)
Forskolin	*Coleus forskohlii*: Ayurvedic use in heart and lung disorders	Adenylyl cyclase activator (from screening of Indian plants)

Product	Source and traditional basis	Outcome
AYUSH-64	Classical antimalarial combination (CCRAS)	Repurposed for mild COVID-19 under AYUSH protocols
Jeevani	*Trichopus zeylanicus* (arogyapacha), used by the **Kani tribe** for stamina	Anti-fatigue product (JNTBGRI) with **benefit sharing** to the Kani tribe
Metformin	*Galega officinalis* (goat's rue, French lilac): folk remedy for diabetes (guanidine)	First-line antidiabetic

1.2 Integrative approaches from the AYUSH perspective

Integrative medicine: combining **evidence-based traditional (AYUSH) and modern medicine** for the patient's benefit.

Integrative approach	Examples
Integration in health services	AYUSH doctors and medicines co-located in district hospitals and primary health centres (**National AYUSH Mission**, NHM); **Ayushman Arogya Mandirs** (AYUSH)
Integrated research	**ICMR-AYUSH** collaborations, **CCRAS**, **CSIR** programmes (e.g. "Golden Triangle Partnership"), **AIIA** (All India Institute of Ayurveda) integrative clinics
Integrated treatment protocols	AYUSH interventions as **add-ons** in diabetes, hypertension, arthritis, cancer supportive care, post-COVID care
Global	**WHO Global Centre for Traditional Medicine** (Jamnagar), **WHO Global Traditional Medicine Strategy 2025-2034**, ICD-11 traditional medicine chapter
Systems approach	Ayurveda's **holistic** view (dosha, prakriti) linked with **systems biology**, **Ayurgenomics** (genomic basis of prakriti) and **network pharmacology**

1.3 Ethnopharmacological approach to bioprospecting

Ethnopharmacology: the scientific study of **biologically active agents used traditionally** by human groups (plants, animals, fungi, minerals): their **ethnobotanical, chemical and pharmacological** basis.

Bioprospecting: the **systematic search** for **useful products** (medicines, agrochemicals, enzymes, cosmetics, foods) from **biological resources and associated traditional**